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Course code: CSA1613

Course Name: Data warehousing and Data Mining.

Assessment Tool - 3

Co-1

Concept Map-1:

Decision Support System (DSS)



Components

- * Data
- * Model
- * User-Interface



Types

- * Data-driven
- * Model-driven
- * Knowledge-driven



Flow

Data → Analysis → Decision.



Role

- * Helps Managers
- * Improves decisions
- * Saves time.



Example.

Bank → loan approval.

Concept Map 2: Operational System (OLTP) vs DSS (OLAP)

OLTP

Daily Transactions.
Current data
Used by employees
Fast Processing

DSS (OLAP)

Data analysis
Historical data
used by managers
Decision Making.

Similarity

Both use the same business data

Interaction

OLTP → Data warehouse → OLAP (DSS)

Examples

- * OLTP: ATM, online Banking.
- * OLAP: Sales Report, Business Analytics.

Concept Map 3:

Architecture of Data warehouse

Data Sources



ETL

- * Extract
- * Transform
- * Load.

Data warehouse

↓
Data Marts

↓
Front-End Tools

- * OLAP
- * Reports
- * Dashboards

Metadata

Stores information about data.

Data Flow

Data Sources → ETL → Data warehouse → Reports.

Concept Map 4-

OLAP, Data cubes & Dimensional Modeling

Data warehouse

↓
Data cube

- * Dimensions
- * Measures

↓
OLAP operations

- * slice
- * Dice
- * Roll-up
- * Drill-down

work

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OLAP

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D

Schemas

* Star schema

* Snowflake schema

↓

Tables

* Fact table

* Dimension Table

↓

Application

Sales Analysis, Business Reports.

Concept Map-5:

Query Processing & BI Tools (KNIME)

Data

↓

Query Processing

* Parsing

* Optimization

* Execution.

↓

BI Tool (KNIME)

↓

work flow

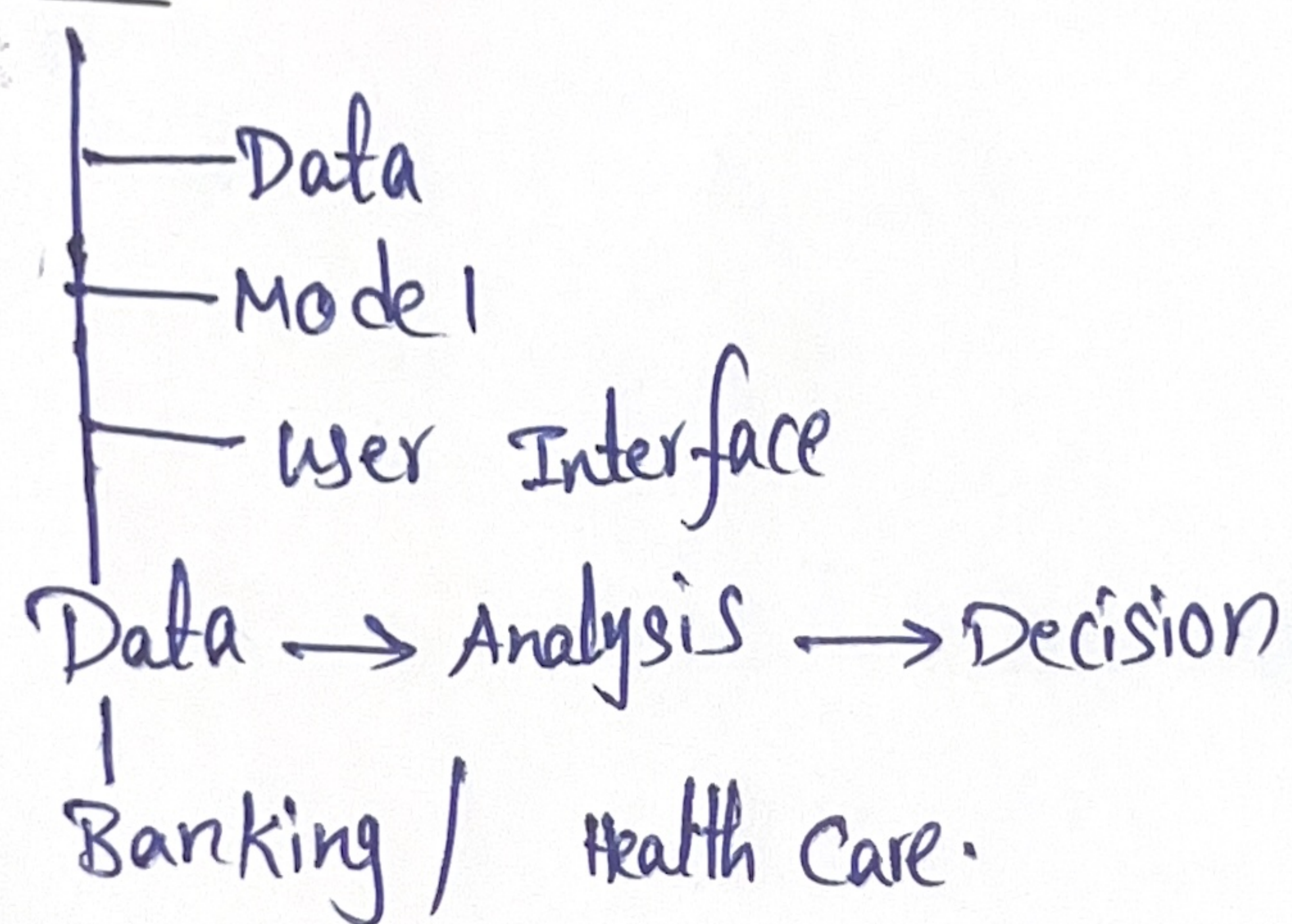
- * Nodes
- * Connections
- ↓
- Visualization
- * Charts
- * Reports
- * Dashboards



OLAP Analysis

- * Slice
- * Dice
- * Roll-up
- * Drill-down.

DSS:



Data warehouse:

Data Sources



ETL



Data warehouse



Data Marts



Reports / OLAP.